Web2Music: Page-Conditioned Music Generation with a Disclosure-Aware Classifier Cascade

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***Abstract*—A browser extension that plays music conditioned on the page a person is reading has to solve three problems: describe the page, keep the page text from leaving the device, and hide a music generator that is slower than browsing. We present Web2Music, a Manifest V3 Chrome extension that addresses all three. The page descriptor takes 86 ms per page without the embedding model (360 ms with the embedding model). A three-stage classifier (keywords, an on-device language-inference model, a hosted large language model) matches the macro-averaged F1 of a keyword-then-LLM baseline on the 255-page development set (0.689) while sending 38% fewer pages off the device; on a held-out test set of [S-E: n pages] pages its macro-F1 is [S-E: test macro-F1]. Playback starts with a short mood-level clip that becomes audible 36 ms after the mood is decided. Music generated for the specific page replaces it only when generation finishes, which takes a median of 13 s for a 15 s clip on a consumer GPU, 62 s on the same machine's CPU and 20 s on an Apple M4; in a browsing simulation with a cold cache the page-level audio is heard on 52% of visits with the GPU, 48% with the M4 and 34% on the CPU (15 s clips, median dwell 30 s). Choosing each loop's cut point by its measured seam lowers the share of seams with an energy jump above 1 dB from 73% to 9% on 176 fresh clips, and the loops play back gaplessly. Audio generated for a page is closer to its intended mood than audio generated for a page of another mood on three objective scorers, but the effect is small; no listening study was run.**

***Index Terms*—generative audio, context-aware computing, browser extensions, zero-shot classification, model cascades, on-device inference, ambient interfaces**

I. Introduction

Background music is chosen once, for a session, and then ignored: a lo-fi playlist runs for four hours across a tax form, a horror wiki and a birthday message without registering that anything changed. Making it context-aware has meant recommending [11] from a library that anticipates every context. Text-conditioned generative audio models remove that requirement (§II): once a description can be derived from what the user is doing, music can be composed for it.

Web2Music does this for the web. Doing it well raises three engineering problems, none of them about audio quality.

1) **The prompt must be derived.** A page is navigation, advertising and colour, not a prompt.
2) **Classification is a privacy decision.** The most accurate classifier is a hosted large language model (LLM), and calling one on every page would turn a music player into a browsing-history channel.
3) **Generation is slower than browsing.** A cold 15 s clip takes 13 s on a laptop GPU, 20 s on an Apple M4 and about a minute on a laptop CPU, against a page load of about two seconds.

This paper contributes one measured answer to each.

1) A four-signal page descriptor (text, sentence embedding, colour, behaviour) that costs 86 ms per page excluding embedding inference and 360 ms with it.

2) A three-stage cascade (keywords, then a zero-shot natural-language-inference model, then an LLM) that treats the share of pages sent off the device (the *off-device rate*) as a measured, adjustable quantity. With an on-device middle stage it matches the keyword-then-LLM baseline's macro-F1 (the mean over categories of each category's F1) on the development set while disclosing 38% fewer pages, and the stage's gate moves the trade-off down to a fully local configuration.
3) Fallback-then-swap playback, which keeps a sound audible 36 ms after the mood is decided whatever the generation time, and loop synthesis that turns a one-shot generator into gapless loops; choosing the cut by measured seam cuts the share of seams with an energy jump above 1 dB from 74% to 12%, and from 73% to 9% on a fresh, seeded set of clips at the shipped crossfade.

Because generation is slow, what a listener hears falls into three *levels of conditioning*. **L1** is a mood-level clip, one of 11 pre-generated clips, one per mood; it is always available. **L2** is a profile-level clip from the cache, generated earlier for a similar music profile (mood, tempo bucket, energy). **L3** is a page-level clip, generated for this page. L1 is audible within 36 ms; L2 and L3 take longer, and Section V-F measures how often a listener reaches them. This paper claims music *conditioned on* the page. It does not claim that listeners find the music fits the page; no listening study was run.

II. Related Work

MusicGen [21] uses a single autoregressive transformer across four EnCodec [22] codebooks with a delay switching between the codebooks so that a single model generates each stream rather than a chain of models. It uses a frozen T5 encoder to condition on text, optionally on melody, and provides checkpoints ranging from 300M to 3.3B parameters. The smallest model can be run on consumer devices. MusicLM [23] generates hierarchically from semantic tokens to the SoundStream acoustic tokens conditioned on the MuLan embeddings of music and text. These generated tokens remain coherent for many minutes. However, model parameters are not made available. AudioLDM [24] uses CLAP embeddings to perform latent diffusion, which allows training on raw audio. Stable Audio [12] uses timing embeddings so a single model can generate audio of arbitrary length. Both models trade autoregressive generation for a complexity that grows with the number of sampling steps. Text-conditioned environmental sounds are generated by AudioGen [25]. All of these models generate a fixed-length segment of audio where the end of the generated audio is not connected to the start.

LoopGen [1] creates a generative model that can be used to generate loops without retraining. LoopGen adapts MAGNeT, a masked non-autoregressive generative model, to generate its own token sequence in a loop. The end of the sequence is generated with the context of the beginning of the sequence, and the join between the end and the beginning is generated rather than later stitched on. While LoopGen leverages the loopable generation from MAGNeT, the loopable generation does not easily translate to autoregressive models. Instead, a loop can be cut after generation at a structural position identified by self-supervised multi-level representations [13] or barwise self-similarity [26]. AffectMachinePop [2] is an expert system which can generate pop music in real time from a desired valence–arousal trajectory [3] without the use of a GPU. This allows for direct manipulation of the perceived affect in the generated music. However, the desired affect must be provided externally.

There have been three relatively recent approaches that create from user context. MetaBGM [14] uses an LLM to convert game and interactive actions into musical descriptions which are then converted into music by a text-to-music system. The music is maintained across scene changes, but the full state is sent to a cloud LLM on each request. Discussion Jockey 2 [4] transcribes speech from an online meeting and then converts it to prompts for MusicGen following the discussion. A user study found a relaxation effect but the full transcript is sent. Context-AI Tunes [5] uses a combination of a user reported stress level and a picture of their current context to create a prompt for relaxation music. The system was tested on 26 users but has active input and the image exits the device. Casper [6] is a browser extension that cleanses user input prior to sending it to a cloud LLM. It uses a local cascade of identifier rules, NER for locations and names and then an LLM to identify sensitive information. Casper cleanses rather than creates and has no exposure rate at the corpus level.

FrugalGPT [15] created the confidence-gated cascade. Queries are attempted using increasingly cheaper LLMs and a learned scorer makes a decision to either use the response or proceed to the next LLM. It was shown that performance comparable to GPT-4 can be achieved with up to 98% less inference cost. Hybrid LLM [16] attempts each query with the LLM predicted to be most appropriate for the query based on difficulty. Up to 40% fewer large models are used while maintaining output quality. Teacher-student caching [27] uses an online small model trained on outputs from an LLM to reduce the number of times the LLM is queried. Agreement-based cascading [7] only progresses to the next model when a group of low cost models do not agree. These systems focus on reducing cost and latency rather than the content seen by the expensive model. WeInfer [8] demonstrates the viability of running an LLM within the browser using WebGPU.

III. Research Gaps and Problem Statement

A. Gaps

Prior work leaves three major research gaps. First, system-level properties have gone unmeasured: evaluation ends at a system's output, typically audio quality. Second, cascades have been optimized for cost efficiency rather than disclosure; no existing cascade permits adjustable exposure. Third, the paucity of generators that produce loops and the missing seams at post-processing have gone unmeasured. Web2Music seeks to close these gaps by systematically deriving music description from any web page passively and attacking all three at once: it enables the measurement of the entire pipeline

from the DOM to audio, treating exposure as an adjustable rate, and ensuring all loops are closed with a measurable seam.

B. Problem statement

Web2Music is judged on four properties. **Latency:** how soon after a mood is decided something is audible, and how soon page-level audio follows. **Disclosure:** how much page text leaves the device; this is measured, not assumed. **Continuity:** clips loop without gaps. **Restraint:** pages with sensitive content produce no audio and send nothing. The zero-shot and LLM providers, and the key proxy in front of them, are designed to receive only the page title, a two-sentence description and the top keywords. The generation service is designed to receive only the music profile (category, mood, tempo and related fields); with a local GPU that channel disappears too. A network audit (Section V-K) checks both statements.

IV. System Architecture

Web2Music is a Manifest V3 [17] Chrome extension plus one HTTP generation service. Fig. 1 follows one page from DOM to speaker in nine steps and marks the two persistent stores: an IndexedDB vector store on the device and a Postgres cache index behind the generation service.

A. Page descriptor

One function assembles the descriptor from four signals: text; a 384-dimensional MiniLM sentence embedding computed in-browser; colour, an area-weighted HSL histogram over up to 800 elements; and behaviour, scroll and cursor velocity; plus a readability score. The embedding is matched by cosine similarity against an IndexedDB store of prior pages to flag revisits, capped at 2,000 entries.

B. Classifier cascade

The classifier maps a descriptor to one of 13 content categories and one of 11 moods, and these select the music profile and the prompt. The shipped default is the full three-stage cascade with the on-device middle stage (configuration A6b in Section V-D). **Keyword stage:** at the keyword level, the keyword map is used to find the category, while moods take the keyword score, colour, and behaviour biases together into account [9]. The cascade also stops at this stage, at zero cost, when confidence is ≥ 0.5. **Zero-shot stage:** the text is scored against thirteen category hypotheses using natural-language inference (NLI) [28]. With $s_{(1)}$ and $s_{(2)}$ the highest and second-highest scores, the stage takes the top label only if

$$s_{(1)} \geq \tau \quad \text{and} \quad s_{(1)} - s_{(2)} \geq \delta, \tag{1}$$

that is, the best label is strong enough and clearly ahead of the runner-up; in any other case it abstains and the page moves to the next stage [7]. The defaults are $\tau = 0.45$ and $\delta = 0.10$. The stage has two interchangeable back ends: an online hosted bart-large-mnli endpoint, and a quantised ONNX nli-deberta-v3-xsmall model that runs in the offscreen document without sending any data. In the ablations the hosted back end is replaced by the same bart-large-mnli weights run locally at full precision (fp32 ONNX in transformers.js, no data leaves the device); the 8-bit quantised export of that model returned near-uniform scores and was not used, and the deployed offscreen model remains the quantised deberta-v3-xsmall. **LLM stage:** the gpt-oss-20b model running on Groq is used, with the temperature set to 0. A second, fully local configuration uses Qwen2.5-7B-Instruct (4-bit, served by Ollama) in the same role; it builds the page profiles for the fidelity study and is reported separately from the Groq results. An eight-second timeout limit is implemented; in case of timeout, the decision would be made on the basis of keywords. The LLM stage can be switched off in the settings; the extension then makes no LLM request at all, which is configuration A6. The chosen mood will stay the same for five seconds in the confidence period before the musical transition to avoid the perceivable fluctuations due to user scrolling. The externally hosted stages are reached through a proxy that holds the API keys, so no credential is embedded in the extension. The share of pages that leave the device is then

$$D = \frac{1}{N} \sum_{i=1}^{N} P(\text{page } i \text{ reaches an off-device stage}), \tag{2}$$

which is the quantity the gate (τ, δ) adjusts. Every stage's output is logged, and the LLM's raw answers are cached, so any configuration can be replayed offline without querying the model again (§V-D).

Sensitive-page check. Before either hosted stage, a term list flags sensitive pages. An optional entailment check then scores the page against three hypotheses: a personal crisis, an academic or news discussion of a difficult topic, and everyday content with no difficult topic. The third hypothesis is what lets an ordinary page (a recipe, a tutorial) score as neither crisis nor discussion. A sensitive page produces no audio and is never sent to the LLM.

From mood to prompt. A committed mood picks a particular row from a per-mood table, including parameters like tempo span, keys, timbre, instruments, reverb, and style. Tempo is calculated using interpolation according to the energy parameter in the chosen range and then tuned with the help of one of nine page-type tuners and seven time-of-day tuners. Key is determined by valence, while darker pages are given more reverb. This profile fills the prompt template ending with loop-based directives (without vocals, without percussion on the loop point).

C. Generation, loop synthesis and caching

The service verifies the query and queries the Postgres cache index (steps 5–7), where the index is exact on the mood, style, and key but also on the energy, valence, arousal, tempo, and duration buckets. If the match was successful, the URL and loop point are returned. In case of a miss, MusicGen-small (300 million parameters, default sampling) is run on one batching worker. The back end is a start-up setting: MusicGen-small (default), MusicGen-medium, or AudioLDM2-music, a latent-diffusion model, so that Section V-H can compare

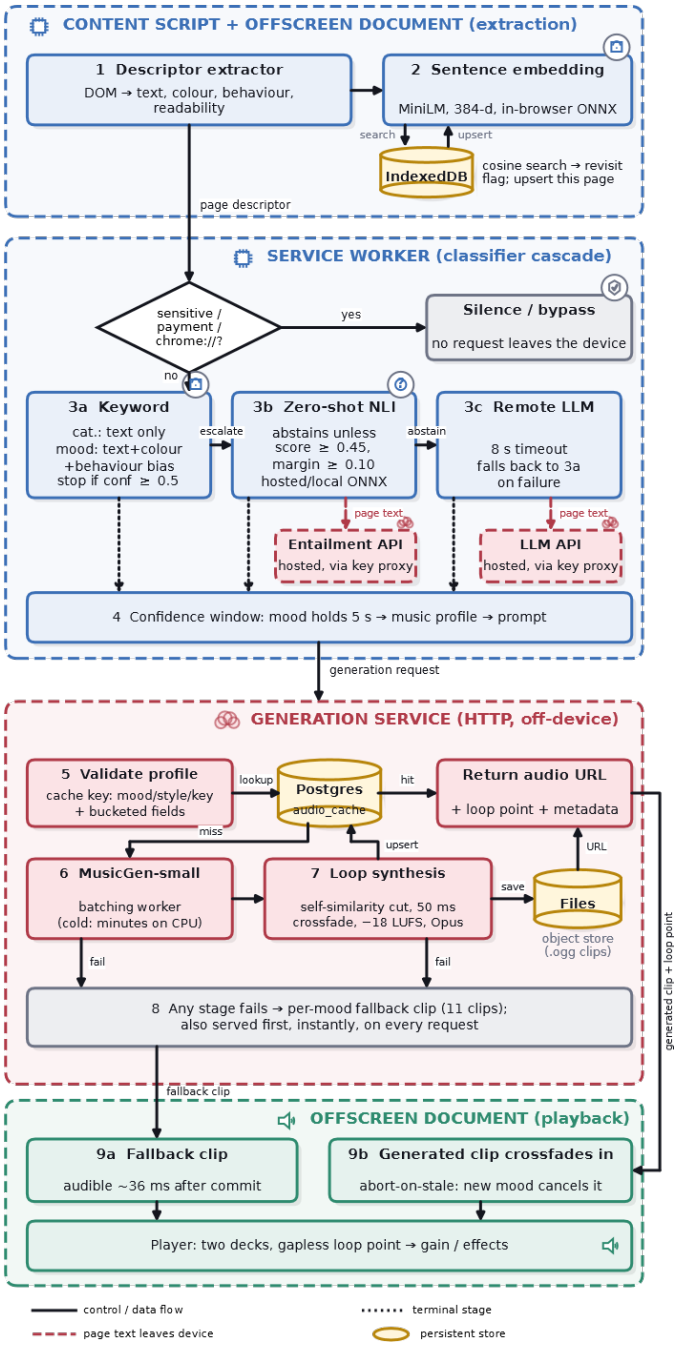


Fig. 1. System architecture of Web2Music.

generators under one pipeline. The worker batches up to four queries within a 150 ms time window, and it generates 50 tokens per second of requested audio that is then processed into an audio clip with index upserting. In case of failure during any step, the system uses a pre-generated audio clip for the particular mood.

Loop synthesis. Chroma descriptors give a self-similarity curve [26] whose peaks, together with the bar boundaries, are the candidate loop points (at least 3 s into the clip). The default selector cuts at each candidate, applies an equal-power crossfade (250 ms in the shipped build), and keeps the candidate whose crossfaded seam is smallest:

$$J = |\Delta E| + \lambda\,|\Delta c|, \tag{3}$$

where ΔE is the energy jump in dB between the end and start of the loop, Δc the jump in spectral centroid in Hz, and $\lambda = 1/5000$ per Hz, so the centroid term only breaks ties. The earlier rule, which cuts at the chroma-similarity maximum, remains available as a switch for comparison (§V-I). Loudness is normalised to -18 LUFS (loudness units relative to full scale) [29] and the clip is exported as Ogg/Opus, which, unlike MP3, adds no padding at every repeat.

D. Playback and restraint

Playback is embedded inside an off-screen document. Looping is performed by two decks using the raw audio buffer inputs alternatively and then applying three gain stages (user volume control, tab reduction, and mood fade), where all the gain stages multiply and not overwrite each other. The fallback-then-swap strategy involves accessing the fallback clip based on mood and the next generation simultaneously, playing back the fallback clip and fading in the new generation based on the equal power curve of the seam for 3 seconds; the previous generation is killed when there is a new mood. **Restraint:** sensitively flagged pages do not have any audio playbacks and are skipped before the start of either of the two hosted stages. When the current tab is generating audio, then the player is muted. On the other hand, background tabs playing audio decrease the gain to 0.1 of the player's volume. URLs of payment sites, banking websites, and internal browser URLs are exempted from the process. The system keeps up to 2,000 telemetry events in a local ring buffer, with each URL replaced by an HMAC-SHA-256 digest under a random per-install key (a plain hash of a URL can be reversed by hashing likely URLs). The buffer is never sent anywhere unless the user exports it. What leaves the device is: in the fully local configuration (A6), only the music profile sent to the generation service; in the default configuration (A6b), additionally the title, description and keywords of pages that the on-device stages could not decide, sent to the LLM through the proxy.

V. Results and Discussion

A. Evaluation design

We report a systems benchmark, a cascade ablation with a held-out test set, a browsing simulation of how often generated audio is heard, objective conditioning-fidelity measures, a loop quality study, and a restraint audit. Percentiles are nearest-rank. Where an interval is given it is a 95% percentile bootstrap over pages or clips (10,000 resamples).

Corpora. The *development set* is 255 real pages in seven slices (general, vocabulary mismatch, non-English, mixed-genre, sensitive, near-miss, bypass), 252 of them category-labelled. Every threshold is chosen on it. The *test set* is **[S-E: n]** further pages captured after the system was frozen, from a URL list fixed before any classifier saw them, and is scored once with the thresholds chosen on development. No URL

TABLE I
HARDWARE (GPU: DRIVER 592.82, CUDA 12.8, PYTORCH 2.11).

Tier	Device	Precision	Used for
CPU laptop	i7-1255U, 16 GB	fp32	stubbed systems benchmark
M4	Apple M4 (MPS)	fp32	cold generation, first prompt study
GPU	RTX 5060 Laptop, 8 GB	fp16	generation, end to end, loops, fidelity
CPU only	i7-14650HX (GPU host)	fp32	CPU-only generation

TABLE II
CASCADE CONFIGURATIONS ON THE 255-PAGE DEVELOPMENT SET. SEVEN CONFIGURATIONS, FROM KEYWORD-ONLY TO LLM-ONLY, SERVE AS THE BASELINES.

	Configuration	Acc.	Macro-F1 [95% CI]	LLM	Off-dev.
A1	keyword only	.190	.315 [.25, .37]	0	0
A2	hosted zero-shot only	.556	.595 [.53, .65]	0	.976
A3	LLM only	.694	.694 [.64, .74]	1	1
A4	keyword → LLM	.679	.689 [.63, .74]	.757	.757
A5	keyword → hosted ZS → LLM	.643	.678 [.62, .73]	.220	.757
A6	keyword → local ZS	.433	.500 [.44, .55]	0	0
A6b	keyword → local ZS → LLM	.671	.689 [.63, .74]	.467	.467

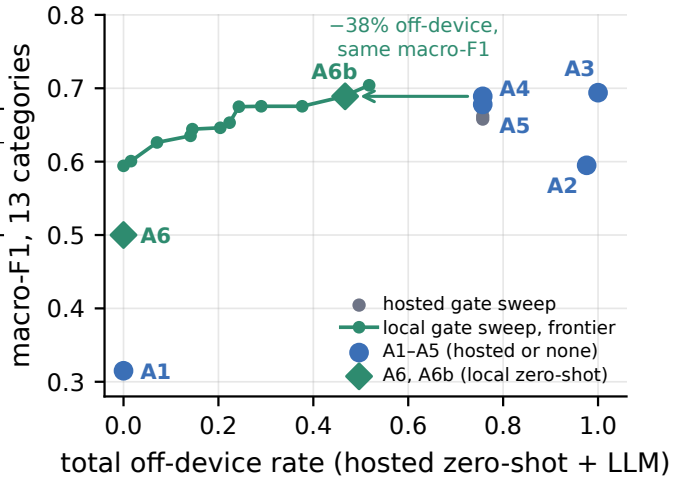


Fig. 2. Macro-F1 versus off-device rate (development set).

TABLE III
HELD-OUT TEST SET: GATES CHOSEN ON THE DEVELOPMENT SET, EACH CONFIGURATION SCORED ONCE. ACCURACY IS SPLIT BY THE STAGE THAT DECIDED.

Configuration	Macro-F1 [95% CI]	Off-dev.	Kw.	ZS	LLM
keyword → LLM (A4)	[S-E]	[S-E]	[S-E]	–	[S-E]
A6b local ZS → LLM	[S-E]	[S-E]	[S-E]	[S-E]	[S-E]
A6 keyword → local ZS	[S-E]	0	[S-E]	[S-E]	–

was suggested by an LLM; the source mix is **[S-E: source mix]**, and **[S-E: n post-cutoff]** pages were published after the LLM's knowledge cutoff, so that contamination can be checked. The *adversarial slice* is 47 hand-written pages (29 sensitive, 18 not) built from the detector's own term lists; it is used only to choose the sensitivity gates. The *black-box set* is **[S-F: n]** real crisis and support pages found by searching for support resources, not by using the term lists.

Labels. Each page carries the category it was collected for. Three of the authors labelled every page independently. For the test set the labelling was blind: the collection category, the system's predictions and the other labellers' answers were hidden, page order was shuffled per labeller, and the guidelines were frozen after a calibration round on development pages and before any test page was labelled. The labellers are the authors, not independent annotators; we state this as a limitation (Section VII). The gold label is the majority of the three, and **[S-E: n ties]** ties were settled by discussion. On the development set the majority resolves for 208 pages and agrees with the collection category on 85%. Only category and binary sensitivity are labelled; mood is not labelled or evaluated. No LLM chose pages, labelled them or judged any output; the LLM stage (model, prompt, temperature 0) was frozen before the test set was opened and its raw answers are cached, so every replay uses the same answers.

Clips and hardware. The first loop study uses the 141 clips in the service's store, 130 generated and 11 per-mood fallbacks; the second generates 220 fresh clips from fixed seeds. The fidelity study generates 702 clips and the generator comparison 55 per generator. Table I lists the hardware. Generation time and the browsing simulation are reported for the GPU, its CPU alone and the M4; the stubbed systems benchmark ran on the CPU laptop.

B. Annotation agreement

Over 255 pages, Krippendorff's α [18], a chance-corrected agreement coefficient, is 0.635 for category, 0.419 for binary sensitivity and 0.195 for mood. Category agreement approaches but does not reach the 0.667 reliability threshold, so we score the cascade on category and read its results as tentative. Mood agreement stays low even after coarsening to valence–arousal quadrants [3] ($\alpha = 0.121$) or NRC VAD norms [10] ($\alpha = 0.300$): the mood of a page is a matter of taste. Web2Music therefore treats mood as a stylistic mapping from page signals rather than a classification target, and mood is not evaluated anywhere in this paper. On the test set, labelled blind under frozen guidelines, α is **[S-E: test alpha category]** for category and **[S-E: test alpha sensitivity]** for binary sensitivity, with per-category values in the supplementary material.

C. Descriptor cost and signal value

On 105 live websites loaded in an automated Chromium browser, descriptor construction takes a median of 86 ms against a 1,989 ms navigation, and boilerplate removal keeps 80.8% of visible text. Embedding inference is stubbed in this benchmark, so 86 ms is the extraction cost excluding the model; with the real MiniLM embedding model, loaded once and warm as in the extension, on 166 sites the median is 360 ms (p95 519 ms), of which the embedding stage is 343 ms (p95 430 ms). Of 204 sites attempted, 38 could not be loaded (DNS or connection errors on infrastructure domains) and are excluded; the embedding failed on none of the 166. Text

TABLE IV
LATENCY BY REQUEST PATH (WALL-CLOCK TIME AT THE CLIENT). "CPU ONLY" IS THE GPU HOST WITH THE GPU DISABLED; NAVIGATION ROWS ARE THE REAL PIPELINE, 110 LOADS EACH, OF WHICH 10 GENERATE.

Path	HW	n	p50	p95
Fallback clip served	CPU laptop	33	7 ms	153 ms
Cache hit, end to end	CPU laptop	93	2.09 s	2.11 s
Cache check, pooled connection	M4	30	8 ms	41 ms
Cold generation, 10 s clip	CPU only	12	38.5 s	39.5 s
Cold generation, 15 s clip	CPU only	12	62.2 s	63.6 s
Cold generation, 28 s clip, one mood	M4	30	72.1 s	105.9 s
Cold generation, 28 s clip, 11 moods	M4	110	72.2 s	119.4 s
Cold generation, 28 s clip	GPU	33	24.3 s	25.0 s
Cold generation, 15 s clip	GPU	33	13.3 s	13.6 s
Navigation to generated audio, A6 (LLM off)	GPU	110	7.7 s	37.6 s
Navigation to generated audio, A6b (local 7B LLM)	GPU	110	7.9 s	64.6 s
Navigation to audible, stubbed	CPU laptop	54	6.86 s	11.28 s

extraction, the colour histogram, and readability take medians of 11.0, 3.9, and 1.5 ms; most of the remainder, and the whole p95 tail, is the vector-store search, which is idle-scheduled. An offline ablation on 255 pages replays the keyword stage with one signal removed: removing colour changes the mood label on 7.5% of pages, and reducing the descriptor to text alone changes 10.2%, the extra 2.7 points coming from the readability term; both shape about a tenth of mood decisions, which, with no reliable mood ground truth (§V-B), shows influence, not benefit.

D. Classification and disclosure

Table II and Fig. 2 give the result. The single-stage baselines bracket the design space: keyword-only (A1) discloses nothing at .190 accuracy, and LLM-only (A3) reaches .694 by disclosing every page. Keyword-to-LLM (A4) comes within .005 of A3's macro-F1 (paired 95% CI $[-.041, +.033]$) while keeping 24% of pages on the device. Every macro-F1 interval is about $\pm.05$ wide, so smaller differences are not resolved by this corpus.

A local middle stage is what makes the cascade a privacy mechanism. With the on-device zero-shot model, A6b matches A4's macro-F1 (.689; paired difference .000, 95% CI $[-.044, +.042]$) while the fraction of pages leaving the device falls from .757 to .467, a 38% reduction. The local model answers in a median of 390 ms against 1.74 s for the hosted one.

The local gate is the disclosure dial. Replaying A6b with the local stage's gate moved traces the green frontier in Fig. 2. Lowering the score threshold from 0.45 to 0.30 cuts off-device from .467 to .290 at macro-F1 .675, and opening the gate entirely keeps every page on the device at .594, level with the hosted zero-shot model alone (A2, .595); A6 at the default gate reaches only .500 because declined pages fall to a default category. On the development set these gate cells locate the trade-off. For the held-out test set, the gates are chosen on development pages only and every configuration is then scored once (Table III): A6b reaches macro-F1 **[S-E: A6b test F1]** against **[S-E: A4 test F1]** for keyword-to-LLM, a paired difference of **[S-E: paired diff]**, with off-device rates of **[S-E: A6b off-device]** and **[S-E: A4 off-device]**. Splitting accuracy by the stage that decided shows how much of each score comes from the LLM: **[S-E: by-stage]**. On the **[S-E: n post-cutoff]** test pages published after the LLM's knowledge cutoff, macro-F1 is **[S-E: post-cutoff F1]** against **[S-E: pre-cutoff F1]** on the rest.

A hosted middle stage only redistributes disclosure. A5 cuts LLM-specific exposure from .757 to .220, but every page it keeps from the LLM reaches the hosted zero-shot endpoint instead, so total off-device stays at .757, as the cascade's structure dictates. A twelve-cell sweep of the hosted gate shows it: macro-F1 stays within .658–.685 and LLM exposure moves over a $5.4\times$ range (.106–.576), while total off-device holds at .756–.757 in every cell. The hosted gate chooses which provider sees a page; only a local backend changes whether one does.

E. Generation latency and fallback-then-swap

Table IV and Fig. 3 show the routes a request can take. The fallback clip is served in 7 ms, so the audible start is never delayed by the generation process. On the Apple M4, all 140 cold generations of 28 seconds complete, with a median of 72 seconds and a real-time factor (RTF, generation seconds per second of audio) of 1.8–2.3 across durations; the generation process itself takes up 71.5 seconds of that median, and every other stage takes under a second. With the GPU disabled, the same machine generates a 10 s clip in a median of 38.5 s and a 15 s clip in 62.2 s (real-time factor 3.9–4.2; 12 requests per length spread over three moods, none timed out), so the fallback clip plays for a minute or more while a cold clip is generated. Cache hits on the CPU laptop took 2.09 seconds, almost all of it spent in the cache lookup, and using pooled Postgres connections lowers that lookup to 8 ms. On the RTX 5060 a cold 15 s clip takes a median of 13.3 s and a 28 s clip 24.3 s (real-time factor 0.87–0.91 end to end across 8–28 s clips, of which the generator call alone is 0.83–0.85; 33 cold requests per length, spread over all 11 moods, none failed or fell back).

The whole pipeline was then measured with the real classifier, embedding model and generator: the built extension loads 11 fixture pages 10 times each (110 loads per setup, of which the first load of each page generates and the rest hit the cache). With the LLM stage off (A6), the first sound is heard a median of 7.7 s after navigation (p95 12.1 s), almost all of it the 5 s confidence window, and the generated clip takes over at p95 37.6 s; the generation round trip is 29.8 s at p95 for the 28 s default clip. With the LLM stage on (A6b) the first sound is heard at 7.9 s (p95 9.2 s) and a classifier decision takes a median of 1.2 s against 0.9 s, but the generated clip arrives later (p95 64.6 s): the LLM in this run is a local 7B model that shares the 8 GB GPU with MusicGen, which roughly doubles generation time, so this row bounds a fully local setup on one GPU rather than a hosted LLM. All 220 loads became audible and none timed out. Concurrent requests do not raise throughput: $k = 2$, 4 and 8 simultaneous 15 s requests finish in 28.3, 54.8 and 111.3 s, within 1–2% of running the same requests one after another (speedup 1.02, 1.01 and $1.01\times$; five

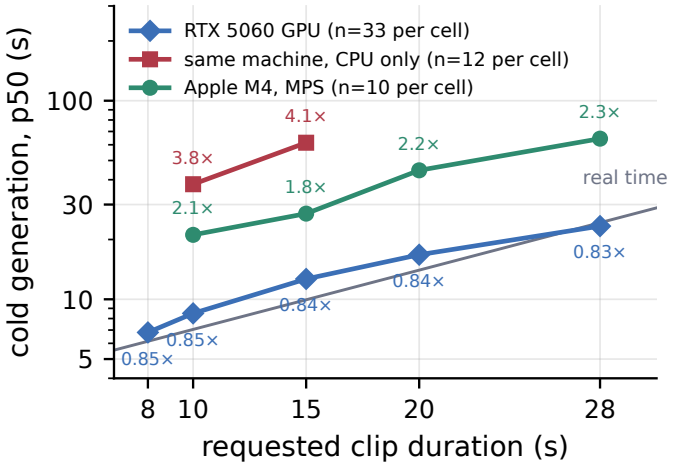


Fig. 3. Cold generation time (p50) versus clip duration on the RTX 5060, on the same machine with the GPU disabled, and on the Apple M4; labels are the real-time factor of the generator call.

repeats per level, rotating moods, none failed), so the worker's batching brings no gain on this GPU and the queueing delay grows linearly with k. On the CPU-only configuration a cold 15 s clip takes about a minute, so generated audio arrives only in longer visits, and for most visits the system behaves as a mood-level player with cache hits (Section V-F).

An automated browser tests the built extension against eleven fixture pages with generation and classification stubbed out: out of 55 page loads, 54 became audible, with a median latency of 6.86 seconds from navigation. The first extraction finishes in 0.86 seconds, and the mood then stays stable throughout the confidence window (median of 6.0 seconds); once the mood commits, the fallback becomes audible in a median of 36 milliseconds (p95: 54 milliseconds), regardless of what the generator is doing. This is the latency property of §III: the remaining latency is the confidence window, which is a design parameter, not model latency. The p95 value (11.28 seconds) is due to one fixture requiring five extraction passes before its mood commits.

F. How often generated audio is heard

The 36 ms figure describes the mood-level clip (L1), not generated audio. Whether a listener ever reaches the page-level clip (L3) depends on how long they stay compared with how long generation takes:

$$P(\mathrm{L3}) = P\big(T_{\mathrm{gen}} + W < T_{\mathrm{dwell}}\big), \tag{4}$$

where T_{gen} is the generation time, $W = 5$ s is the confidence window before a mood is committed, and T_{dwell} is the time on the page. We estimate it by simulating browsing sessions. Moods follow a Markov chain fitted to the corpus mood frequencies; dwell times are Weibull-distributed [19] with the median swept from 10 to 120 s and a longer median for long-form pages; generation times are drawn from the measured distributions of each tier and clip length (none of the requests timed out); there is one worker, and generation is cancelled when the next page needs a different clip. L1 is the mood clip, L2 a cache hit and L3 a fresh generation that finishes before the visit ends. Cache entries are keyed on the exact profile (mood, tempo, key, energy and the style string), and the cache starts empty for each simulated session. The service's 99-entry pre-warm grid is credited only with the hit rate measured on real profiles: replaying the profile builder's keys for the 235 corpus pages, none falls on a grid key (75 distinct keys occur in all), so L2 arises only from an earlier visit to the same key within a session. Mood frequencies are those of the corpus (curious 36%, focused 18%, neutral 12%, calm 2%).

TABLE V
SIMULATED SHARE OF VISITS AND OF LISTENING TIME AT EACH LEVEL OF CONDITIONING (COLD PER-SESSION CACHE, MEASURED GRID HIT OF 0, MEDIAN DWELL 30 S), IN PERCENT; THE REMAINING 19% OF VISITS END BEFORE A MOOD IS COMMITTED. LONG-FORM AND OTHER COLUMNS GIVE THE L3 SHARE OF VISITS.

Tier	Clip	Visits L1/L2/L3	Listening L3	Long-form L3	Other L3
GPU	15 s	14/15/52	75%	62%	49%
GPU	28 s	21/14/46	72%	59%	42%
CPU	15 s	37/10/34	64%	50%	29%
M4	15 s	19/14/48	73%	61%	44%

The simulation says how often generated audio *could* be heard under stated assumptions; it is not a measurement of listeners. On the GPU, page-level audio is heard on 52% of visits and accounts for 75% of listening time at a 15 s clip; on the M4 the figures are 48% and 73%, and on the CPU 34% and 64%, so generator speed matters less than dwell time and the confidence window. The share rises with dwell time (GPU, 15 s: 37%, 52% and 64% of visits at median dwell 10, 30 and 120 s) and is higher on long-form pages (52%, 62% and 68%). Two bounds frame these figures. If half of the pages did hit the pre-warm grid, which we do not observe, the GPU share would fall to 29% of visits at 30 s, because a grid hit is served as L2. If the cache persisted across sessions, then since the corpus has only 75 distinct keys nearly every committed visit would be served from the cache after the first sessions (81% of visits are L2 at 30 s); this is a trivial upper bound on cache use, not an estimate of it. The M4 row rests on the three-request-per-length duration sweep.

G. Conditioning fidelity

With no listening study, we ask a narrower, objective question: does audio generated for a page agree with the mood the system intended for it, more than audio generated for a page of a different mood does? The cascade (with the local Qwen2.5-7B LLM stage) builds a music profile for each of the 234 non-sensitive corpus pages; we generate 702 clips (234 pages × 3 seeds) from these profiles on the GPU and compare each with a *shuffled* control, a clip from a page whose mood is at least two steps away in valence–arousal space, plus a page-agnostic fixed-prompt control. Four non-generative scorers are used: Essentia valence/arousal regressors and the MTG-Jamendo mood tagger; LAION-CLAP, which retrieves a page's own clip among 10 candidates from the page's title and lead text (chance is 10%); and Audiobox Aesthetics for quality. No LLM or audio-language model judges any output.

TABLE VI
CONDITIONING FIDELITY, MATCHED VERSUS SHUFFLED CLIPS (PAIRED OVER PAGES; WILCOXON AND BOOTSTRAP CI).

Measure	Matched	Matched − shuffled [95% CI]
Valence Spearman ρ	.40	–
Arousal Spearman ρ	.51	–
Valence–arousal distance to target (closer)	–	+.050 [.026, .076]
Intended mood in top-5 tags	59.9% [56.2, 63.4]	+7.5 pts [1.3, 13.8]
CLAP page-text similarity	–	+.023 [.014, .033]
CLAP top-1 retrieval (chance 10%)	12.5% [10.1, 15.0]	–

TABLE VII
GENERATOR COMPARISON (15 S REQUESTS, RTX 5060; 55 CLIPS PER GENERATOR, PAIRED BY MOOD AND SEED). RTF IS END TO END. FAD IS THE FRÉCHET DISTANCE IN LAION-CLAP AUDIO SPACE TO THE 11 SHIPPED FALLBACK CLIPS (LOWER IS CLOSER); PQ IS AUDIOBOX PRODUCTION QUALITY.

Generator	RTF	Seam $\lvert\Delta E\rvert$ p50	Retention p50	FAD (CLAP) ↓	Aesthetics PQ
MusicGen-small	0.89	0.11 dB	.548	0.33	7.34
MusicGen-medium	1.61	0.12 dB	.536	0.37	7.41
AudioLDM2-music	0.51	0.05 dB	.602	0.68	6.87

Matched clips are closer to the intended mood than shuffled ones on all three scorers (paired Wilcoxon $p = .010$, .029 and 4×10^{-5}; the mixed-model intercepts, which also account for the three seeds per page, give $p = 2 \times 10^{-6}$, .004 and 3×10^{-8}), and valence and arousal track the intended values ($\rho = .40$ and .51, $p < 10^{-27}$). The effects are small, however. The tagger's hit rate for a shuffled mood is already 52%, and CLAP retrieves a page's own clip among ten candidates in only 12.5% of cases, barely above the 10% chance, so the audio is shifted toward the intended mood on average but a page's clip cannot be picked out from its text. The matched clips score 7.03 [6.97, 7.09] on Audiobox Production Quality, against 7.33 for the 11 shipped fallback clips and 6.95 for the fixed-prompt control (3 clips). These are proxies for conditioning fidelity, not for perceived fit. CLAP is trained on music captions, not web pages; the mapping from moods to tags is an assumption fixed before scoring; and agreement between a prompt and its audio shows only that the generator follows its prompt, which is why page text against audio is the primary CLAP measure.

H. Generator comparison

The pipeline is generator-agnostic. Table VII runs the same 11 moods and 5 fixed seeds through MusicGen-small (the default), MusicGen-medium and AudioLDM2-music, a latent-diffusion model, on the GPU. It shows the effect of scale and of a different model family, and every other stage is unchanged.

MusicGen-medium is the slowest (RTF 1.61, so a 15 s clip is not ready before it would have finished playing) and is no better than the default: its seam differs from MusicGen-small's by -0.03 dB [-0.08, 0.01], and its PQ and FAD are within the small model's confidence interval or marginally worse. AudioLDM2-music is the fastest (RTF 0.51) with the smallest seam jump, but it sits farthest from the fallback clips in CLAP space (FAD 0.68) and has the lowest PQ (6.87 [6.68, 7.05]). Without a listening study we cannot say which timbre is preferred; the data support the default only on speed and seam quality.

TABLE VIII
LOOP-SEAM QUALITY BY CUT METHOD (113 USABLE CLIPS OF THE 141 IN THE STORE).

Cut point, crossfade	$\lvert\Delta E\rvert$ p50	$\lvert\Delta E\rvert$ p95	>1 dB	$\lvert\Delta c\rvert$ p50
Chroma, 50 ms (orig. default)	2.44 dB	10.29 dB	74%	405 Hz
Chroma, 100 ms	2.03 dB	7.87 dB	81%	314 Hz
Chroma, 250 ms	2.18 dB	7.41 dB	71%	190 Hz
Chroma, 500 ms	1.82 dB	7.56 dB	67%	133 Hz
Measured seam, 50 ms	0.14 dB	1.54 dB	12%	288 Hz

I. Loop quality

Of the 141 delivered clips, all fall within their duration bounds, and loudness lands at -18.09 LUFS against the -18.0 target. The 50 ms crossfade reduces the median seam energy discontinuity from 3.73 to 2.64 dB and improves 58.9% of seams. Delivered loops keep a median 50.3% of the requested duration; since the loop repeats gaplessly, this sets the loop period rather than the playback length, so a 28 s request repeats about every 14 s.

Table VIII and Fig. 4 compare the two levers on the same 113 clips. Increasing the crossfade tenfold, from 50 to 500 ms, lowers the median energy jump by only 25%, and 67% of seams remain above 1 dB. The measured-seam selector, which keeps the 50 ms crossfade and chooses among the detector's own candidate points, brings the share of seams above 1 dB down to 12% and above 3 dB to 1% (from 74% and 36%), and the median drops from 2.44 to 0.14 dB. It makes the seam better on 102 of 113 clips, worse on two, and preserves internal duration retention (.695 vs. .690). The selector minimises this same energy jump together with a small spectral-centroid term, so these numbers show how far the candidates let the jump fall, not that listeners hear it; the 500 ms crossfade still gives the smallest centroid jump. The shipped build uses a 250 ms crossfade with the measured-seam selector; this table compares the levers on the 50 ms setting they were measured at.

Two further checks replace listener judgement. First, a fresh study of 220 clips (11 moods × 20 fixed seeds) compares the two rules on the shipped crossfade. Of the 220, 176 are scoreable: 39 have a tail shorter than 500 ms, too short for the 250 ms crossfade, and 5 have no detectable loop point. On these 176 the median $\lvert\Delta E\rvert$ is 2.08 dB [1.69, 2.33] for chroma similarity and 0.16 dB [0.12, 0.23] for the measured seam, and the share above 1 dB falls from 72.7% [65.9, 79.0] to 8.5% [4.5, 12.5] (McNemar $p = 2 \times 10^{-34}$; the 145 clips that fit the full crossfade sweep give 71.7% and 6.2%). Second, the loops are *gapless*: rendering each clip through the browser's own decoder and a looping buffer source for 10 cycles, 30 of 30 clips show no inserted silence and are sample-identical from cycle to cycle, so the Ogg/Opus container adds no padding at repeats. Relative to each clip's own variation, the seam's

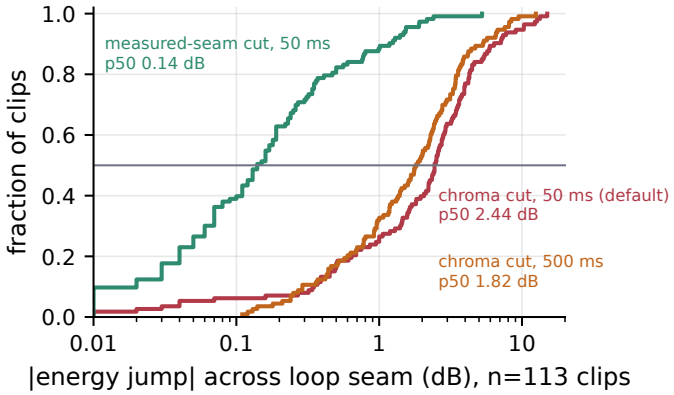


Fig. 4. Energy jump across loop seams, by cut method (the 113 clips of Table VIII).

TABLE IX
PROMPT COMPARISON.

Prompt	Chars	$\lvert\Delta E\rvert$	$\lvert\Delta c\rvert$	Retention
P1 bare mood word	6	3.75 dB	565 Hz	.527
P2 service template	175	2.63 dB	509 Hz	.444
P3 engineered (default)	312	4.35 dB	316 Hz	.393

spectral-centroid change lies within the clip's 95th percentile of interior change in 91.4% [87.7, 95.0] of the 220 clips. The energy jump is within its 95th percentile in all 220, which is expected because the selector minimises that jump, so we do not count it as evidence. Gapless playback is shown; a seam being *inaudible* is not claimed.

J. Prompt ablation

The default prompt is an engineered description of 312 characters. On the Apple M4, we tested it against the bare mood word (3–9 characters) and the service's template (about 175 characters), keeping mood, duration and seed constant (Table IX).

On the M4 the two shorter prompts gave a smaller energy jump and longer loops than the engineered one (15 generations per prompt, no statistical testing), which is an indication, not a ranking. The fixed-seed rerun on the GPU (11 moods $\times$ 5 seeds per prompt, 165 clips, paired by mood and seed) does not reproduce a difference: the median seam $\lvert\Delta E\rvert$ is 0.14 dB for the bare word, 0.11 dB for the template and 0.14 dB for the engineered prompt, the paired difference engineered minus template is 0.00 dB $[-0.10, 0.04]$ (Wilcoxon $p = 0.88$), and bare minus template is -0.05 dB $[-0.11, 0.05]$ ($p = 0.34$). All three take the same time to generate (13.2–13.3 s median) and keep a median 47–55% of the requested duration. The prompt therefore has no measurable effect on loop quality here, and the same seeds are reported with every result. Whether the engineered prompt's timbre is preferred is untested.

K. Restraint

All fourteen executable restraint policies pass their tests: silence on sensitive pages [20], muting and ducking around existing audio, bypass of payment and browser-internal pages, hashed-URL telemetry, and seven unit cases. The adversarial slice then attacks the sensitivity predicate.

The term lists are precise, producing only three false positives, all near-miss academic pages such as a grief-counselling degree. However, a white-box slice built from the lists' own vocabulary evades them on 19 of 29 sensitive pages, through out-of-list terms (7), euphemisms (4), non-English text (4), and single ambiguous terms (4). Re-scoring with the entailment model detects all 29, including the four non-English pages, with scores between 0.83 and 0.92 (Table X). This run used the hosted bart-large-mnli backend, which receives every page without a hard-severe term; in this configuration, the check itself therefore discloses the pages it is meant to protect. The disclosure property holds only with the on-device backend. The hosted two-hypothesis check also overtriggers: its eleven false positives include all six plainly benign pages, such as a Rust tutorial, because the prompt offers only crisis or reference discussion and no "unrelated" option. The shipped check therefore adds that third hypothesis and runs on-device. With its gates chosen on the hand-written slice only, it detects **[S-F: recall [Wilson CI]]** of the **[S-F: n]** real black-box pages (**[S-F: non-English recall]** of the non-English ones) and raises a false alarm on **[S-F: FPR [Wilson CI]]** of **[S-F: n benign]** benign pages, with a median latency of **[S-F: ms]** ms (Table X). All detector claims come from the black-box and benign sets, not from the hand-written slice.

Network audit. To check the disclosure claims on the wire, we recorded every request made by the service worker, the offscreen document, the content scripts and the pages while the built extension browsed 12 fixture pages, one per mood plus a crisis-support page, three times each. In the fully local configuration (A6, the LLM stage switched off), no request body contained page text (the only traffic was the generation service, 850 bytes per page). In the default configuration (A6b), page text left the device on 30 of the 36 visits (83%), all to the LLM proxy and nowhere else (5.9 kB per page to the proxy and 0.8 kB to the generation service). The fixtures were written to exercise the cascade and are not representative of browsing, so this rate is not the disclosure rate (47% of the 255 development pages in the replayed cascade); what the audit establishes is the destination of the text. No telemetry left the browser, and every request to the generation service carried the music profile and nothing else. The generation service therefore sees the category and mood of each page; with a local GPU it is local too.

VI. CONCLUSION

Web2Music plays background music conditioned on the page being read. Each pipeline step is assessed against the properties of Section III. Running the middle classifier stage on the device keeps the keyword-to-LLM macro-F1 on the development set and cuts off-device page transfers by 38%; the gate sets how much further disclosure is reduced, and the held-out result is in Table III. On the wire, the fully local configuration sends no page text, and the default sends it only to the LLM proxy. A mood-level clip is audible 36 ms after

TABLE X
SENSITIVE-PAGE DETECTION: THE HAND-WRITTEN SLICE (29 SENSITIVE, 18 NOT), THEN REAL BLACK-BOX PAGES AND BENIGN PAGES.

Detector	Missed	False alarms	Non-English caught
Term lists (default)	19/29	3/18	0/4
Term lists + NLI (hosted)	0/29	11/18	4/4
Term lists + NLI (on-device, 3-way)	[S-F]	[S-F]	[S-F]
Black-box set and benign pages (Wilson 95% intervals)			
Term lists (default)	[S-F]	[S-F]	[S-F]
Term lists + NLI (on-device, 3-way)	[S-F]	[S-F]	[S-F]

the mood decision whatever the generation time; with a laptop GPU the page-level clip is heard on about half of the simulated visits, and on a CPU on about a third. The generated audio leans toward the intended mood, but only slightly on objective scorers. Choosing the loop point by measured seam removes most large energy jumps at the loop, and the loops are gapless.

VII. LIMITATIONS AND FUTURE WORK

No perceptual evaluation. We did not run a listening study, so we make no claim that listeners find the music fits the page, that the loops sound seamless, or that the 14 s loop period does not cause fatigue. The protocol and instruments for a within-subjects study (page-matched against shuffled audio; forced choice, Likert ratings, dwell, skip and mute) are released with the code. The conditioning-fidelity measures of Section V-G are proxies and inherit the biases of the models behind them: CLAP is trained on music captions, not web pages, and the mood-to-tag mapping is an assumption. The measured effects are small (CLAP retrieval of 12.5% against 10% chance), so the fidelity result shows a tendency, not that a page's music is recognisably its own.

Author annotation. The three labellers are authors. Labelling was blind and the guidelines were frozen first, but this is not independent annotation, and agreement between authors who share a design may overstate reliability.

Simulation assumptions. How often generated audio is heard depends on dwell times that we model rather than measure: the Weibull shape and the long-form multiplier. The share of pages whose profile is in the pre-warm grid is measured on the corpus (0 of 235), but the corpus is not a sample of real browsing. The sweep over the median dwell shows the sensitivity, not the truth for any particular user.

Measurement scope. The end-to-end runs and the network audit use 11 and 12 fixture pages, so they establish timings and where text goes, not rates over real browsing. The A6b end-to-end run used a local 7B LLM on the same GPU as the generator, which slows generation; a hosted LLM would not.

Other limits. The held-out test set is small, so macro-F1 intervals are about $\pm.05$ wide and smaller differences are not resolved. The LLM stage in the main configuration is a hosted model; its answers are cached for replay but its behaviour can change under the same model name, and the local Qwen2.5-7B configuration gives different answers and is not interchangeable with it. The hardware is one laptop GPU (also measured CPU-only), one Apple M4 and one CPU laptop. Future work includes conditioning the generator to produce loop-closed content, independent annotators and a listening study, and applying the same pipeline to other continuous contexts such as documents and code editors.

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